# SNO Tenant Attachment to the EVPN Fabric — Architecture Design

Oct 15, 2026 · @nimuk

## Summary and decision

Each SNO cluster attaches to a tenant VPC in exactly one of two modes, chosen per cluster by what is physically cabled. Point-to-point VXLAN from an SNO to a single border leaf is retired, because that tunnel dies with its leaf and nothing re-homes it.

- **Mode A, VLAN trunk.** The SNO has a tagged trunk to each border leaf. The SNO is not a VTEP. An active-backup bond moves the guest MACs to the surviving leaf.
- **Mode B, host VTEP.** The SNO shares only the IP underlay with the leaves. The SNO host runs a kernel VXLAN device per VNI and FRR EVPN, as a client of both route reflectors. Multus only plugs the guest into a bridge.

Both modes can serve the same VPC from different clusters. One cluster never uses both modes for the same VPC, because the same MACs would then appear on two paths.

1. Gateway failover time is designed: BFD on every EVPN session, datacenter timers, and next-hop tracking where the underlay supplies VTEP /32s.
2. Losing both route reflectors stops all cross-node traffic once sessions drop. EVPN-learned MACs do not age out; they are withdrawn.
3. SNO hosts now originate EVPN routes, so the reflectors filter what each SNO may advertise.
4. DHCP gets real high availability instead of a single speaker leaf.
5. MTU, ARP suppression, inbound (corp to VM) paths and live migration are specified.
6. OpenShift implementation and its conflicts (frr-k8s, MetalLB) are covered.

## Scope, assumptions and addressing

The design covers L2 tenant VPCs (L2VNIs) with a centralised anycast gateway on the border leaves. Symmetric IRB and L3VNIs on the SNOs are out of scope.

Assumptions:

- The underlay is routed, owned by the network team, with jumbo MTU and UDP/4789 plus TCP/179 permitted. Mode B widens that permit (see Control-plane security).
- BL1, BL2, RR1 and RR2 are physical servers running FRR in AS 65001, iBGP L2VPN EVPN. Both leaves have eBGP to both corp routers.
- Both leaves advertise the tenant /24 to corp. Corp may ECMP inbound traffic across them.
- Guests use KubeVirt bridge binding on a Multus bridge NAD with no IPAM. The pod network stays management only.

Lab addressing used throughout:

| Item | Value |
|---|---|
| BL1 / BL2 VTEP | 10.99.0.11 / 10.99.0.12 |
| BL1 / BL2 router ID | 10.255.0.1 / 10.255.0.2 (confirm VTEP and router ID are intentionally different) |
| SNO1 / SNO2 host VTEP (Mode B) | 10.99.0.21 / 10.99.0.22 |
| Tenant VNI / VLAN | 10310 / 310 |
| Tenant subnet | 10.50.10.0/24 |
| Anycast gateway | 10.50.10.1, MAC 00:00:5e:00:01:01 |
| Example guests | VM-A 10.50.10.25 on SNO1, VM-B 10.50.10.26 on SNO2 |

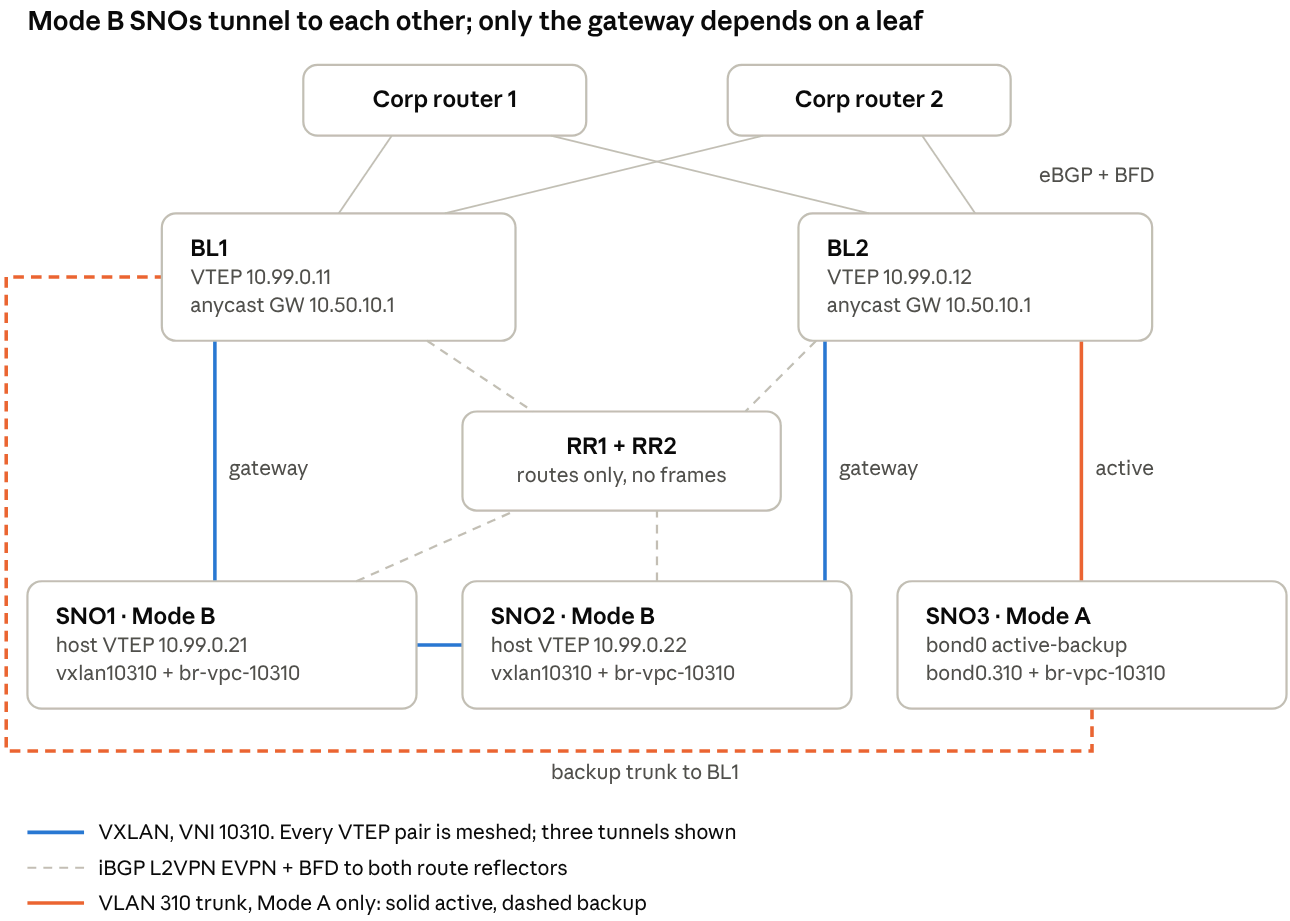
## Fabric roles

The reflectors carry routes only; every tenant frame flows between VTEPs, and the leaves are the only path to corp.

| Component | Control plane | Data plane | Holds state for |
|---|---|---|---|
| BL1, BL2 | iBGP EVPN to RR1 and RR2; eBGP to both corp routers; BFD on all sessions | VTEP per VNI; anycast SVI 10.50.10.1; local bridge ports for Mode A trunks | Gateway, corp routes, DHCP, Mode A guest MACs |

| Component | Control plane | Data plane | Holds state for |
|---|---|---|---|
| RR1, RR2 | Reflect Type-2 and Type-3 to all clients; enforce per-client policy | None | All EVPN routes |
| SNO, Mode A | None for tenants | Active-backup bond into br-vpc-<vni> via a tagged sub-interface | Nothing; MACs live on the leaves |
| SNO, Mode B | FRR iBGP EVPN to RR1 and RR2, advertise-all-vni, BFD | Kernel vxlan<vni> enslaved to br-vpc-<vni> | Its own guest MACs |
| Multus / KubeVirt | None | Guest tap on br-vpc-<vni> | Nothing |
| Corp routers | eBGP to both leaves | Route to and from the tenant /24 | Corp routes |

In Mode B, two guests on the same SNO bridge locally and never encapsulate. In Mode A, two guests whose bonds are active on the same leaf bridge on that leaf.




Attachment architecture · both modes on one VNI

SNO1 and SNO2 reach each other directly. For Mode B the leaves carry only gateway and corp traffic; for Mode A they carry both trunks.

## Mode A: VLAN trunk attachment

Use Mode A only where the SNO has one tagged trunk to BL1 and one to BL2, both carrying every tenant VLAN the cluster uses.

### SNO host

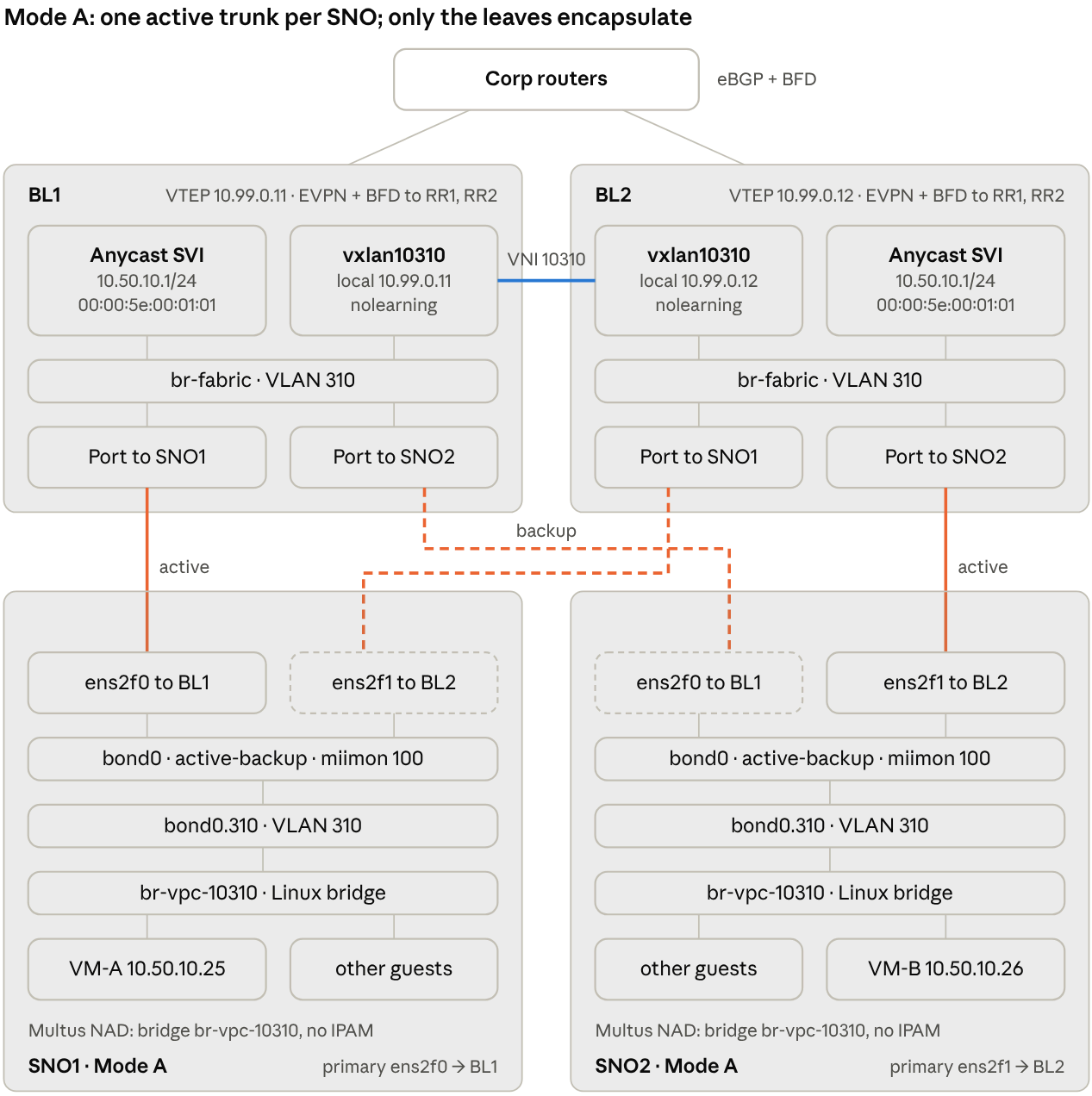
- bond0 in active-backup mode over the two trunk NICs. Not LACP: BL1 and BL2 are separate servers with no MLAG or EVPN multihoming between them.
- `miimon 100`, `fail_over_mac none`, `primary` set per cluster so clusters spread across both leaves.
- bond0.310 enslaved to br-vpc-10310. The bridge NAD is identical to Mode B.

### Border leaves

- The SNO-facing port (or its VLAN 310 sub-interface) joins the leaf's br-vpc-10310 next to vxlan10310.
- The leaf learns guest MACs locally and advertises them as Type-2 from its own VTEP.
- **Link-state tracking.** If a leaf loses its underlay or both RR sessions, it admin-downs its SNO-facing ports so the bond moves. miimon alone only detects a dead link, not a live link into a broken leaf.

An ARP-monitored bond is the alternative to link-state tracking. It needs a host IP on the trunk, which tenant VLANs do not give the host, so it is not the baseline.

# Mode A in detail




Mode A detail · host stack, leaf bridge and trunks

Each SNO has one tagged trunk to each leaf, and the bond forwards only on its primary member. Clusters alternate primaries so both leaves carry traffic. All VXLAN work happens between the leaves.

**SNO host configuration.** Example NNCP for SNO1. SNO2 is identical except `primary: ens2f1`. Interface names are examples.

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: sno1-tenant-trunk
spec:
  nodeSelector:
    kubernetes.io/hostname: sno1
  desiredState:
    interfaces:
    - name: bond0
      type: bond
      state: up
      ipv4: {enabled: false}
      link-aggregation:
        mode: active-backup
        port: [ens2f0, ens2f1]
        options:
          miimon: "100"
          primary: ens2f0
          primary_reselect: failure
          fail_over_mac: none
          all_slaves_active: "0"
    - name: bond0.310
      type: vlan
      state: up
      ipv4: {enabled: false}
      vlan: {base-iface: bond0, id: 310}
    - name: br-vpc-10310
      type: linux-bridge
      state: up
      ipv4: {enabled: false}
      bridge:
        options: {stp: {enabled: false}}
        port:
        - name: bond0.310
```

| Bond setting | Value | Why |
|---|---|---|
| mode | active-backup | The leaves are separate servers with no MLAG or EVPN multihoming |
| miimon | 100 | Link loss detected in about 100 ms |
| primary | ens2f0 on odd SNOs, ens2f1 on even | Spreads clusters across both leaves |
| primary_reselect | failure | No automatic fail-back when a leaf returns, so guests take one MAC move, not two. Rebalance in a change window |
| fail_over_mac | none | The bond MAC stays the same on either member |
| all_slaves_active | 0 | The backup member drops what it receives, so floods from the backup leaf never re-enter the bridge |

The NAD and VM spec are the same as Mode B: `type: bridge`, no IPAM, `macspoofchk: true`.

**Border leaf configuration.** br-fabric is VLAN-aware; vxlan10310 and the anycast SVI already exist from the fabric plan. Each SNO trunk port is added as a tagged member of VLAN 310.

```
# BL1: SNO1 trunk on enp65s0f0, SNO2 trunk on enp65s0f1 (example names)
ip link set enp65s0f0 master br-fabric up
bridge vlan del dev enp65s0f0 vid 1
bridge vlan add dev enp65s0f0 vid 310
bridge link set dev enp65s0f0 guard on   # BPDU guard: drop the port if a
loop appears
# already present:
#   vxlan10310 in br-fabric, vid 310 pvid untagged, nolearning
#   br-fabric.310  10.50.10.1/24  MAC 00:00:5e:00:01:01
```

Because the leaf owns the SVI, it learns guest IPs as well as MACs and advertises MAC+IP Type-2 routes. The other leaf can then answer ARP for those guests without flooding.

**Link-state tracking on each leaf.** A small systemd service, driven by FRR BGP and BFD state, admin-downs the leaf's SNO-facing ports when either is true:

- No EVPN session to either route reflector is established.
- The leaf has lost its underlay uplinks.

It must never trigger on loss of the other leaf alone, or both leaves would shut their ports at once. Ports come back only after EVPN is up to both reflectors, Type-3 routes are present for every VNI, and a 60 s hold-down has passed. Confirm that each NIC model drops carrier on `ip link set down`; if it keeps the link up, the bond never sees the failure.

**Packet paths, steady state** (SNO1 primary on BL1, SNO2 primary on BL2)

1. **VM-A to VM-B.** Tagged VLAN 310 to BL1. BL1 holds VM-B's MAC as a Type-2 at 10.99.0.12, encapsulates to BL2, which sends it down SNO2's active port.
2. **VM-A to a guest on the same SNO.** Bridged on br-vpc-10310; it never leaves the host.
3. **VM-A to the gateway.** BL1's SVI answers and routes to corp. North-south traffic splits across leaves by bond primary.
4. **Corp to VM-A, arriving on BL2.** BL2 routes onto VLAN 310, finds VM-A's MAC at BL1 and encapsulates to 10.99.0.11.
5. **Floods from VM-A.** BL1 sends them to its local trunk ports and replicates to BL2. BL2 sends them down SNO2's active port and SNO1's backup port, where the bond drops them.

**Failover sequences.** Times are design targets, to be measured in the lab.

| Event | Time after failure | What happens |
|---|---|---|
| BL1 loses power | 0 | Carrier drops on SNO1's ens2f0 (active) and SNO2's ens2f0 (backup). Frames inside BL1 are lost |
| | ~100 ms | miimon moves SNO1's bond to ens2f1. SNO2 does not move |
| | ~100 ms to ~1 s | BL2 still holds BL1's Type-2 routes for SNO1's guests and tunnels to the dead 10.99.0.11. Guests that transmit are learned on BL2 immediately |

| Event | Time after failure | What happens |
|---|---|---|
| | ~1 s | BFD on the reflectors drops BL1 and its routes are withdrawn. BL2 floods unknown unicast, reaching silent SNO1 guests through the new active port |
| | as each guest transmits | BL2 advertises the MAC with a higher MAC-mobility sequence |
| BL1 loses the fabric, links stay up | under 2 s | Link-state tracking downs BL1's SNO ports; then as for power loss |
| BL1 returns | after hold-down | SNO ports stay down until EVPN is up and 60 s pass. With primary_reselect failure, bonds stay on BL2 until a planned rebalance |
| One trunk link fails | ~100 ms | Only that SNO moves. If it was the backup member, nothing moves |
| VXLAN path between the leaves fails, both up | immediate | Bridging within each leaf continues; cross-leaf east-west fails. Runbook: down one leaf's SNO ports so all bonds converge on the other |
| SNO host fails | node loss | Its guests stop. The leaf ages out their MACs and withdraws their Type-2 routes |

**Closing the silent-guest window.** A bond-failover hook on the SNO (NetworkManager dispatcher) can send one RARP per guest MAC learned on br-vpc-<vni>, the same way a migrated VM announces itself. That removes the wait for guests to transmit. It is optional; validate it before relying on it.

**Cost of Mode A.** When a leaf fails, the survivor carries all east-west traffic for every cabled cluster plus all north-south. That concentration is why Mode B is preferred wherever the trunks do not already exist.

**Future option.** FRR supports EVPN multihoming (ESI, all-active LACP across two leaves). It would replace active-backup, use both links, and remove the link-state-tracking dependency. Treat it as a later phase.

# Mode B: host VTEP attachment

Use Mode B wherever the SNO shares only an IP underlay with the leaves. The SNO host becomes a full EVPN VTEP, so no single leaf is on the path between SNOs.

### SNO host, outside the CNI

- One kernel `vxlan<vni>` per VPC: `id <vni>`, `local` set to the SNO's VTEP address, `dstport 4789`, `nolearning`, `neigh_suppress on`, enslaved to `br-vpc-<vni>`.
- FRR with `advertise-all-vni`, iBGP L2VPN EVPN to RR1 and RR2, BFD on both sessions, `timers bgp 3 9`.
- No static remote VTEPs. Flood lists come from Type-3 routes; unicast remotes come from Type-2.

### Multus and KubeVirt

```
{"cniVersion":"0.3.1","name":"vpc-10310","type":"bridge","bridge":"br-vpc-10310","macspoofchk":true}
```

The tenant interface is `multus.networkName` with `bridge: {}`. `macspoofchk` stops a guest from sending as another MAC, which would otherwise be advertised into EVPN by its host.

**Border leaves.** Unchanged from the fabric plan: `vxlan<vni>`, anycast SVI and `advertise-all-vni`. Add `advertise-svi-ip` (or `advertise-default-gw`) so the gateway MAC and IP reach every SNO as a Type-2 MAC+IP route.

**Route reflectors.** Every Mode B SNO becomes a client of both. They still forward no tenant frames.

### Packet paths in steady state

1. VM-A to VM-B: one tunnel, SNO1 (10.99.0.21) to SNO2 (10.99.0.22). The leaves are not on the path.
2. VM-A to its gateway: the gateway ARP is answered locally on SNO1 by ARP suppression, from the leaves' MAC+IP route. The IP packet is tunnelled to the leaf whose gateway Type-2 SNO1 selected.
3. Both leaves advertise the same gateway MAC with sequence 0. The expected tie-break selects the lower VTEP address, so all SNOs normally send north-south traffic to BL1. Confirm this in the lab and size each leaf for 100% of north-south.

4. Corp to VM-A: corp may deliver to either leaf. The leaf tunnels to 10.99.0.21 if it holds VM-A's MAC+IP; otherwise it ARPs, and that ARP is flooded to every VTEP in the VNI (see ARP suppression below).

**Adding a VPC** is one bridge, one `vxlan<vni>` and one NAD per SNO, plus the existing leaf devices. No remote list is edited; each SNO advertises its own Type-3.

## Gateway and convergence design

Gateway failover takes as long as the fabric takes to withdraw the dead leaf's routes. With FRR default timers (keepalive 60s, hold 180s) and no BFD, that is up to three minutes of north-south blackhole. The design brings it to seconds.

**Why it is slow by default.** A Mode B SNO does not peer with the leaves. It learns that BL1 is gone only when a reflector's session to BL1 times out and the reflector withdraws BL1's routes. Until then, every SNO that selected BL1 for the gateway MAC keeps tunnelling to 10.99.0.11.

**Mechanisms, in order of reliance**

1. **BFD on every EVPN session** (RR to BL, RR to SNO), for example 300 ms x 3. A reflector detects a dead leaf in about 1 s and withdraws its routes to all clients.
2. **Datacenter BGP timers** (`timers bgp 3 9`) as the backstop when BFD is not running.
3. **Next-hop tracking**, where the underlay gives the SNO a specific /32 for each leaf VTEP (for example, SNO eBGP to the underlay). When 10.99.0.11/32 disappears, the SNO invalidates every route with that next hop, even if BL1's BGP sessions are still up. Leave `ip nht resolve-via-default` off; with it on, a default route keeps a dead VTEP looking reachable.
4. **Gateway route marking.** Leaves advertise the gateway with the default-gateway extended community, so it is not treated as a moving MAC and does not trip duplicate-address detection.

**Target, to be measured in the lab:** under 3 s from leaf failure to gateway traffic flowing via BL2.

**Corp isolation on a leaf.** A leaf that loses both corp sessions still answers as the gateway and drops the traffic. Fix one of two ways:

- An iBGP IPv4 session between BL1 and BL2 over the underlay, so an isolated leaf forwards corp-bound traffic to its peer.

- Or conditional advertisement: when the corp default route is absent, the leaf withdraws its gateway Type-2 and stops originating the tenant /24.

**Considered and not chosen as baseline: anycast VTEP.** Both leaves would share a second VTEP address advertised into the underlay, giving ECMP north-south and underlay-speed failover. It breaks Mode A on the same VNI: a guest MAC behind BL1's trunk would be advertised from the shared address, and traffic landing on BL2 would have nowhere to go without a peer link. Revisit it only for VNIs where every attachment is Mode B.

## Control-plane security and route policy

Mode B makes every SNO host an EVPN speaker, so a compromised or misconfigured SNO could hijack the gateway MAC or inject routes into another tenant's VNI. The reflectors enforce what each SNO may advertise.

**Inbound policy on RR1 and RR2, per SNO peer**

- Permit only route types 2 and 3.
- Permit only the route targets of the VNIs assigned to that cluster (from NetBox), deny everything else.
- Deny any Type-2 carrying the gateway MAC 00:00:5e:00:01:01 or a gateway IP.
- Deny routes with a non-zero ESI or a default-gateway community.
- Deny routes whose originator IP is not that SNO's VTEP address.
- `maximum-prefix` per peer, sized to that cluster's expected guest count plus headroom, with warning-only first.

**Session protection.** MD5 on every RR session today; move to TCP-AO when FRR supports it.

**Underlay filtering, a change for the network team**

| Flow | Before (P2P) | Mode B |
|---|---|---|
| UDP/4789 | SNO to BL | Every VTEP to every VTEP: SNO to SNO, SNO to BL, BL to BL |
| TCP/179 | SNO to BL | SNO to RR1 and RR2 |
| UDP/4784 (multihop BFD) | None | SNO to RR, BL to RR |

VXLAN carries no authentication. Restrict UDP/4789 to the known VTEP address list so nothing else on the underlay can inject frames into a VNI.

**Guest protection.** `macspoofchk` on every NAD (Mode B section) stops a guest from claiming another MAC, which its host would otherwise advertise.

## DHCP, ARP suppression, MTU and live migration

**DHCP.** A single speaker leaf is a single point of failure, and two independent servers on one scope can hand out the same address. Pick one:

- **Relay to corporate DHCP** (preferred where a central service exists). Both leaves relay from the anycast SVI. The reply to giaddr 10.50.10.1 can return through either leaf, and both can deliver it. The server sees two copies of each request; that is harmless for one server.
- **Kea HA hot-standby on BL1 and BL2**, leases synchronised over the management network. A DISCOVER is BUM, so both leaves receive it; only the active answers until its partner fails.

The NAD stays IPAM-less in both modes.

**ARP suppression.** `neigh_suppress on` on each SNO vxlan device answers ARP locally when a MAC+IP Type-2 exists. The gateway is always covered, because the leaves advertise its MAC+IP. For guest IPs it works only if the SNO originates MAC+IP routes, which needs zebra to see guest neighbours on br-vpc-<vni>. Validate that in the lab. Until it is proven, assume guest-to-guest and leaf-to-guest ARP is flooded to every VTEP.

**MTU**

| Layer | MTU |
|---|---|
| Underlay | 9000 (confirmed jumbo) |
| vxlan<vni>, br-vpc-<vni> | 8950 |
| Guest nNIC (default) | 1500 |
| Guest nNIC (east-west jumbo, by exception) | up to 8950, only where the corp path handles PMTUD |

**Live migration (Mode B).** After a VM moves from SNO1 to SNO2, QEMU announces the guest MAC. SNO2 learns it on the new tap and advertises a Type-2 with a higher sequence number; SNO1 removes its local entry when the old tap goes. Every VTEP repoints in one update, with no cable or bond involved. Migration traffic itself uses a dedicated migration L2VNI.

FRR's duplicate-address detection freezes a MAC after 5 moves in 180 s by default. Raise `dup-addr-detection max-moves` if VMs are migrated back and forth during testing.

Migration between a Mode A and a Mode B cluster works the same way: the leaf or the SNO that learns the MAC advertises it with the higher sequence.

## Failover conditions

No single device failure stops tenant traffic in either mode, provided BFD is on every EVPN session. Losing both reflectors stops all cross-node traffic in both modes. In all cases, packets already in flight to a failed device are lost.

| Condition | Detected by | Mode A (trunk) | Mode B (host VTEP) | Recovery driver |
|---|---|---|---|---|
| BL1 down | Link loss on trunks; BFD on RR to BL1 | Bonds active on BL1 move to BL2. All cabled east-west and north-south now on BL2 | VM to VM unaffected (SNO to SNO). Gateway repoints to BL2 after BL1's Type-2 is withdrawn | Bond: miimon (~100 ms). Gateway: BFD + withdraw (target < 3 s) |

| Condition | Detected by | Mode A (trunk) | Mode B (host VTEP) | Recovery driver |
|---|---|---|---|---|
| BL1 up but lost underlay or both RR sessions | Link-state tracking on BL1; BFD on RRs | BL1 shuts SNO ports; bonds move to BL2 | Same as BL1 down for the gateway. NHT speeds it up where VTEP /32s are routed | Link-state tracking; BFD |
| BL1 lost both corp sessions | BFD on corp eBGP | Traffic still reaches BL1. Forwarded via BL2 over BL-BL iBGP, or BL1 withdraws the gateway | Same | BL-BL IPv4 iBGP or conditional advertisement |
| BL1 to BL2 underlay path down, SNOs reach both | BFD between VTEPs where configured; otherwise none | Cross-leaf east-west fails; same-leaf bridging continues. Recover by shutting one leaf's SNO ports | VM to VM unaffected. Each leaf serves the gateway to SNOs that reach it. Route exchange via RRs is unaffected unless RRs depend on that path | Underlay repair |
| One RR down | BFD on all sessions to it | No impact | No impact; the other RR carries all routes | None needed |

| Condition | Detected by | Mode A (trunk) | Mode B (host VTEP) | Recovery driver |
|---|---|---|---|---|
| Both RRs down | All EVPN sessions drop | Same-leaf bridging continues. Cross-leaf traffic stops once routes are withdrawn | Same-SNO bridging continues. All SNO to SNO and SNO to leaf traffic stops once routes are withdrawn | RR restore. Graceful restart extends stale routes for planned restarts only |
| One trunk link down | miimon | Bond moves to the other leaf | Not applicable | Bond |
| SNO1 cannot reach SNO2 in the underlay | None in the overlay | Not applicable | VM to VM between them stops; gateway still works. No cable to fall back to | Underlay repair |
| SNO1 cannot reach BL1, can reach BL2 | NHT if /32s routed; otherwise none | Not applicable | Gateway traffic to BL1 blackholes unless SNO1 selected BL2 or NHT withdraws BL1 | NHT on routed VTEP /32s |
| SNO node down | BFD on RR to SNO; node loss | Its guests are down. Leaves age out its MACs | Its guests are down. Its Type-2 and Type-3 routes are withdrawn | OCP / KubeVirt restart elsewhere if storage allows |
| One corp router down | BFD on corp eBGP | Leaves use the other corp router | Same | Corp ECMP |

| Condition | Detected by | Mode A (trunk) | Mode B (host VTEP) | Recovery driver |
|---|---|---|---|---|
| VM live-migrated SNO1 to SNO2 | QEMU MAC announcement | Leaf relearns MAC on the new port | SNO2 advertises Type-2 with a higher sequence | MAC mobility |

Why both RRs down is worse than the earlier drafts said: EVPN remote MACs do not age out. They are removed when the session that carried them drops. With BFD, that happens within about a second. Put the two reflectors in separate failure domains (power, rack, hypervisor if virtual in the lab), and consider a third.

## Scaling characteristics

Mode B scales by adding VTEPs, not cables. Its cost is head-end replication of BUM traffic, which grows linearly with the number of VTEPs in a VNI.

| Dimension | Mode A | Mode B |
|---|---|---|
| Adding a cluster | Two cables and two leaf ports | One RR client, no cabling |
| East-west after a leaf failure | All on the surviving leaf | Stays SNO to SNO |
| North-south | One leaf at a time per bond | One leaf for all SNOs (lowest VTEP wins) |
| BUM copies per flood | Leaf-to-leaf only | One per other VTEP in the VNI (N SNOs + 2 leaves − 1) |
| Route count on RRs | Type-2 per guest MAC, Type-3 per leaf per VNI | Type-2 per guest MAC (or MAC+IP), Type-3 per VTEP per VNI |
| Per-VPC config on each SNO | bond VLAN sub-interface, bridge, NAD | vxlan device, bridge, NAD |

Levers as VPC and cluster counts grow:

- ARP suppression with MAC+IP routes cuts the largest BUM source.

- FRR's single VXLAN device mode with a VLAN-aware bridge replaces one vxlan device and one bridge per VPC once VPC count per SNO reaches tens.
- Each leaf must be sized for 100% of north-south, since one leaf carries it in steady state and after failure.

## OpenShift implementation

Both modes are host configuration plus a bridge NAD; neither changes OVN-Kubernetes or the pod network.

**Host interfaces**

- Mode A: a NodeNetworkConfigurationPolicy (NMState) for the bond, VLAN sub-interfaces and Linux bridges.
- Mode B: an NNCP for the Linux bridges and vxlan devices. If NMState cannot set `nolearning` and `neigh_suppress`, apply them with a MachineConfig systemd unit or from the FRR pod's init container.

**FRR on the SNO host (Mode B)**

- A privileged, hostNetwork DaemonSet with NET_ADMIN and NET_RAW, using the RHEL FRR build. Configuration rendered from NetBox through the existing Ansible/Jinja2 pipeline.
- Alternative: a MachineConfig-managed systemd unit running FRR under podman. It survives cluster API outages but is harder to update.

**Conflicts to check per cluster**

- **MetalLB / frr-k8s.** Both run FRR on the node and bind TCP/179. A cluster cannot run them alongside a second FRR. Either do not use MetalLB BGP mode on Mode B clusters, or put the EVPN config into frr-k8s if its raw-config path is supported for your OCP version.
- **OVN-K hybrid overlay** uses UDP/4789. Confirm it is disabled.
- **VTEP local address.** Use a stable host address. If it sits on br-ex, the encapsulated traffic shares the node's external interface; size and QoS accordingly.